# Reinforcement Learning for Scenic Photo Parameter Adjustment with TOPIQ Reward

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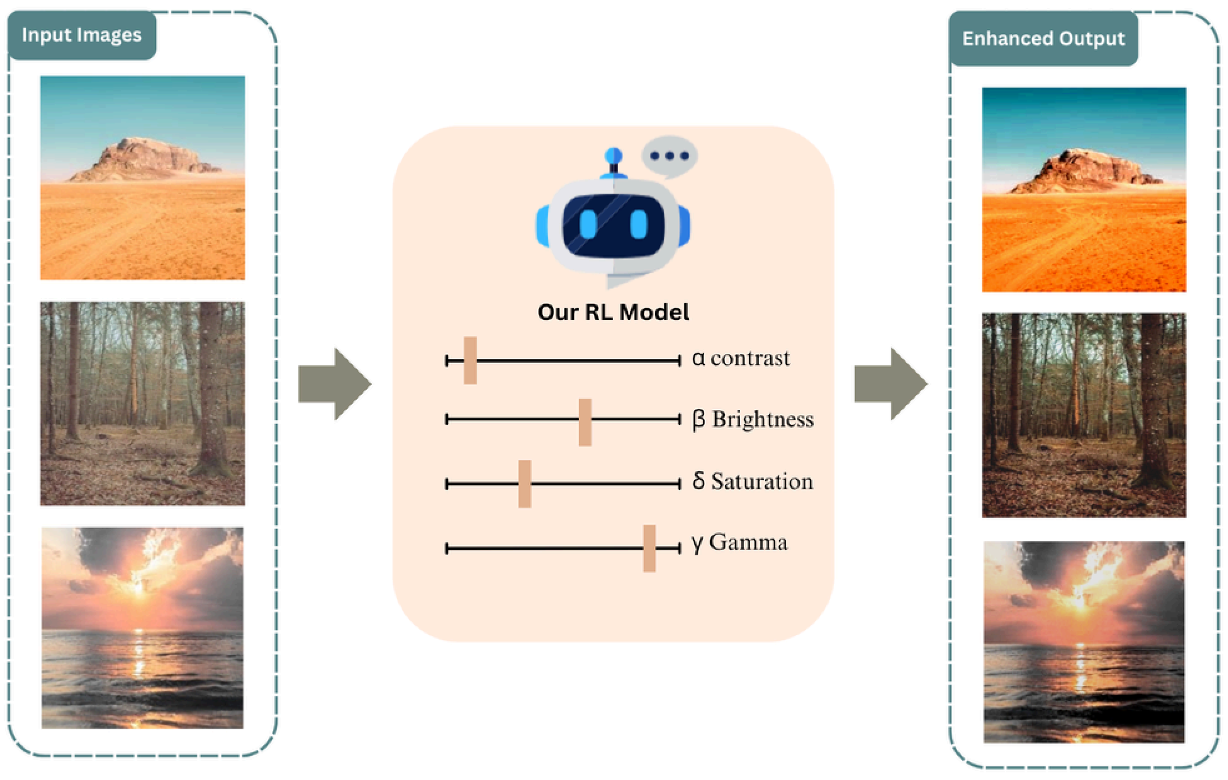
## Task



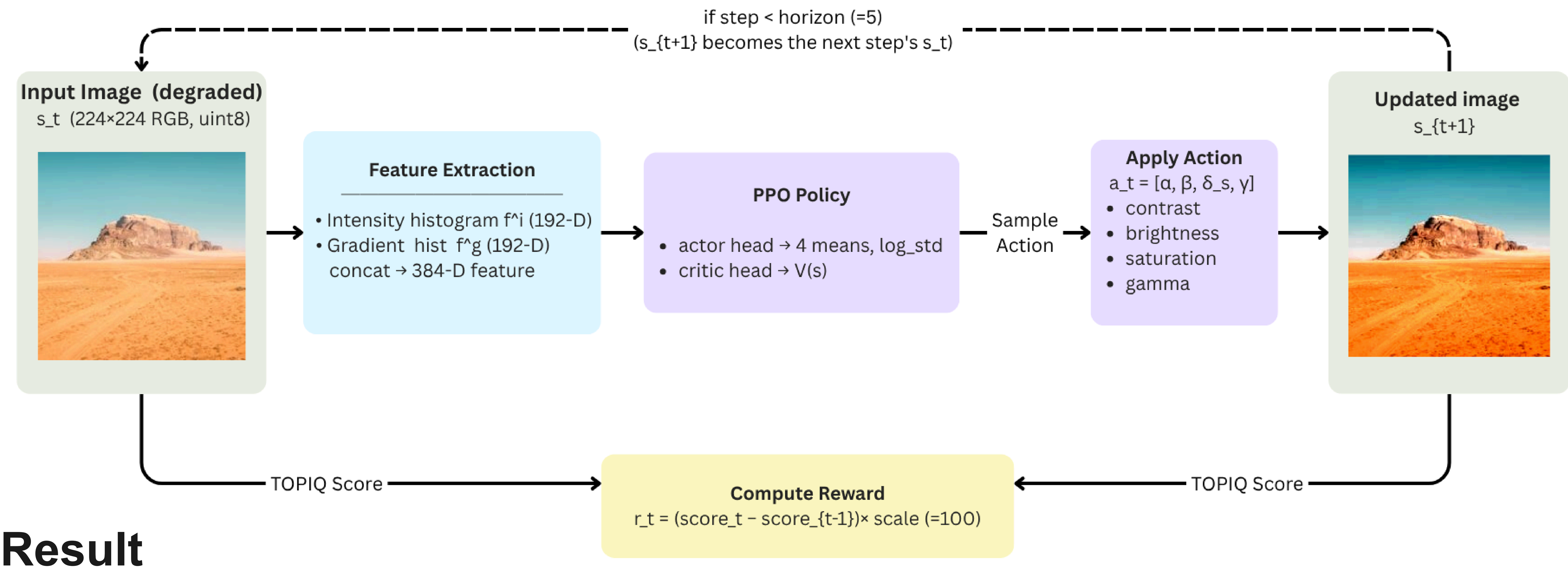

## Motivation

| Method | Requires GT | Method Type | Input Type |
|---|---|---|---|
| DRL-ISP | Yes | ML | RGB images |
| Star Enhancer | Yes | RL | RAW images |
| **Ours** | **No** | **RL** | **RGB images** |

- **No ground truth required.** TOPIQ is a no-reference quality metric, so our policy learns from the image alone, no labeled pairs or reference styles to collect.
- **Reinforcement learning instead of supervised mapping.** PPO lets the agent discover enhancement trajectories that maximize perceived quality, rather than imitating one fixed answer per image.
- **RGB input instead of RAW input.** Most mobile photos are stored as RGB images, while RAW data is often unavailable and cannot be reliably recovered from JPEG.

## Method

One episode (horizon = 5 steps)




## Result

| Method | Input TOPIQ | Output TOPIQ | Improvement |
|---|---|---|---|
| DRL-ISP Baseline | 59.5785 | 45.2433 | -14.3352 |
| Star Enhancer Baseline | 59.5785 | 58.2696 | -1.3089 |
| **Ours (PPO + TOPIQ Reward)** | **59.5785** | **61.1498** | **+1.5713** |

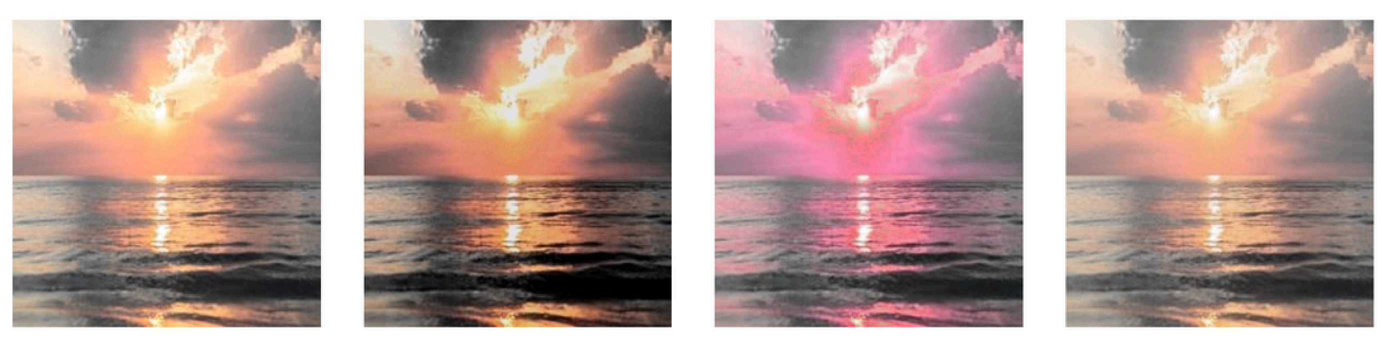
Example 1: Contrast Degradation

Input Ours DRL-ISP Star Enhancer

Example 2: Brightness Degradation

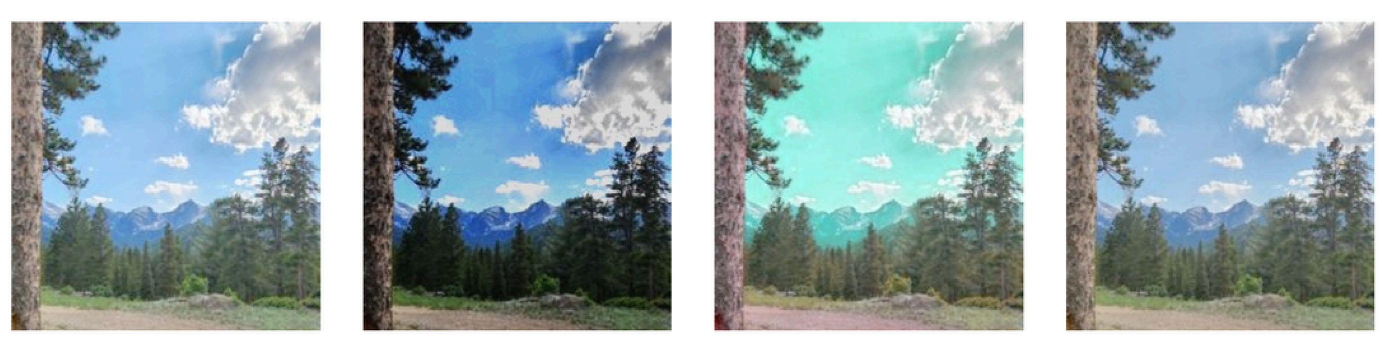
Input Ours DRL-ISP Star Enhancer

Example 3: Saturation Degradation



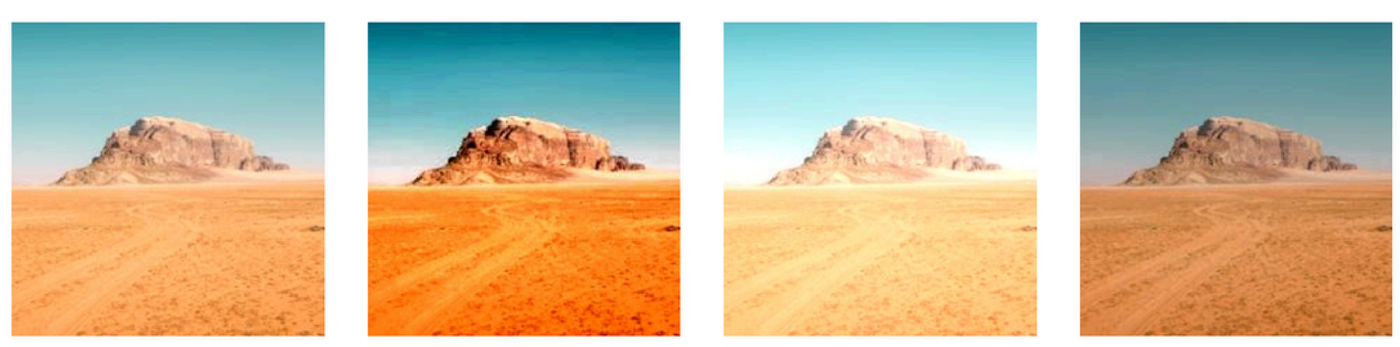
Input Ours DRL-ISP Star Enhancer

Example 4: Gamma Degradation



Input Ours DRL-ISP Star Enhancer

Example 5: All-Parameter Degradation



Input Ours DRL-ISP Star Enhancer

[1] Song, Y., He, H., Qian, H., & Chen, K. (2021). StarEnhancer: Learning real-time and style-aware image enhancement. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 13534-13543).
[2] Shin, U., Lee, K., & Kweon, I. S. (2022). DRL-ISP: Multi-objective camera ISP with deep reinforcement learning. arXiv preprint arXiv:2207.03081.
[3] Chen, C., Mo, J., Hou, J., Wu, H., Liao, L., Sun, W., Yan, Q., & Lin, W. (2024). TOPIQ: A top-down approach from semantics to distortions for image quality assessment. IEEE Transactions on Image Processing, 33, 2404–2418.